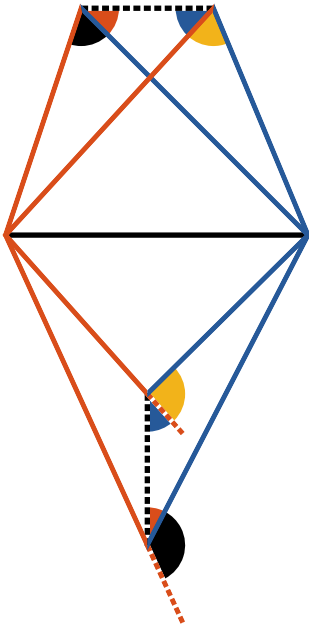




Book I. Prop. VII. Theor.

On the same base (—), and on the same side of it there cannot be two triangles having their conterminous sides (— and —, — and —) at both extremities of the base, equal to each other.

When two triangles stand on the same base, and on the same side of it, the vertex of the one shall either fall outside of the other triangle, or within it; or, lastly, on one of its sides.

If it be possible let the two triangles be constructed so that

$$\left\{ \begin{array}{l} \text{—} = \text{—} \\ \text{—} = \text{—} \end{array} \right\}, \quad \text{then draw}$$
 and,

$$\left. \begin{array}{l} \blacktriangle = \triangle \text{ (pr. I.5)} \\ \therefore \blacktriangle < \triangle \text{ and} \\ \therefore \blacktriangle < \triangle \end{array} \right\} \text{ which is absurd,}$$
 but (pr. I.5) $\blacktriangle = \triangle$

therefore the two triangles cannot have their conterminous sides equal at both extremities of the base.

Q. E. D.